CHAPTER ONE: INTRODUCTION TO ANALYTICS APPLICATIONS

Making data tell a story

Let's watch this:

- Telling Stories with Data - method …
- Hans Rosling's 200 Countries, 20…

Telling Stories with Data - method 2 (The Discovery Journey method)

The first video features Paul Smith discussing an alternative, narrative-driven approach to presenting data called the "Discovery Journey" method. Unlike traditional business presentations that state a recommendation upfront and follow it with supporting evidence, this method walks the audience through the analyst's own process of discovery. To illustrate, the speaker shares a real-world example from his time working for a disposable diaper manufacturer. He explains how he led his leadership team through 40 years of sales and profit data, gradually guiding them to realize for themselves that their market had actually matured back in 1983. By structuring the presentation like a story—complete with context, conflict, and a surprising resolution—he allowed the audience to actively struggle with the data and draw the conclusions themselves. This method relies on creating an emotional "aha" moment for the audience, which makes them much more likely to adopt the recommendations because they feel like they discovered the answer on their own.

Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BBC

The second video is a highly visual and famous presentation by statistician and global health expert Hans Rosling. Using animated, augmented-reality data, Rosling plots 120,000 individual numbers to track the progress of 200 countries over two centuries. He plots life expectancy (health) against income per person (wealth), showing how the world looked in 1810 when almost all nations were categorized as "poor and sick". As the timeline moves forward, Rosling highlights the physical impact of major historical events on the graph, such as the Industrial Revolution, the First World War, the Spanish flu, and the Great Depression. By the end of the animation, he illustrates a converging modern world where the significant historical gap between Western countries and the rest of the world is closing, with many nations moving toward the "rich and healthy" corner. He also breaks down large countries like China into individual provinces to demonstrate that vast economic and health disparities can still exist within a single nation today.



The situation involves the floods getting worse from time to time to the point announcements from barangay and LGUs are made, prompting rescue operations of 3 rescue vehicles on 5 areas. → ???

Using analytical reasoning without data but we would still rely on data.

This is not yet a "data problem." It is first a human and a decision problem.

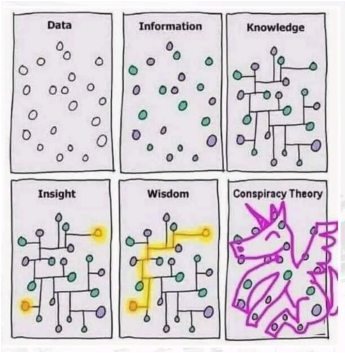
- **Decision** means a choice among possible courses of action made under certain objectives and constraints.
- **Uncertainty** means that the decision-maker does not yet have complete knowledge about the situation, outcomes, or consequences.

- **Analytics begins when a decision meets uncertainty.**



Analytics can be applied to anything or everything.

Analytics can be the answer.



What Do We Need to Know?

1. **Data:** symbols
2. **Information:** data that are processed to be useful; provides answers to "who", "what", "where", and "when" questions
3. **Knowledge:** application of data and information; answers "how" questions
4. **Insights:** The appreciation of "why"
5. **Wisdom:** Evaluated insights.

Decision problem: which of them is the best possible solution.

From Data to Insight

- **Analytical Result** is an output directly produced by an analytical procedure, computation, model, or visualization.
- **Finding** is an analytical result judged meaningful or relevant enough to report.
- **Insight** is a meaningful interpretation of one or more findings that improves understanding of a problem, opportunity, or decision.

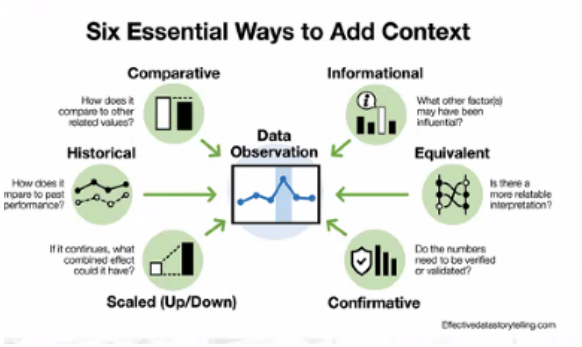


Insight is the most meaningful of them all because it is the most actionable.

Context changes everything

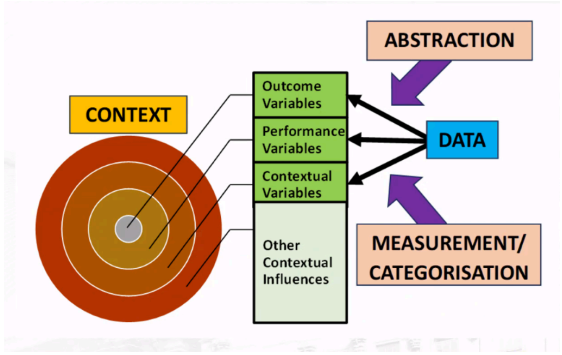
The same number can tell different stories.

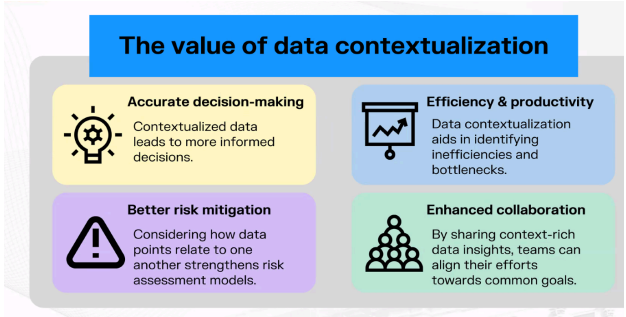
- **Context** refers to the surrounding conditions that influence how data, findings, and analytical results should be interpreted.
- Context may include time, place, people, policy, culture, weather, operations, constraints, and domain knowledge.
- **Contextualization:** Provides background information or context to help the audience understand the significance of the data and its relevance to the larger narrative.



The Relationship Between Context and Data

All variables in a dataset necessarily display what is called “communality”, that is, shared information reflecting the influence of their common context. There will always be some degree of contextual association between variables which makes it difficult to isolate the shape and size of the direct relationship between two variables.





What, then, is Data Context?

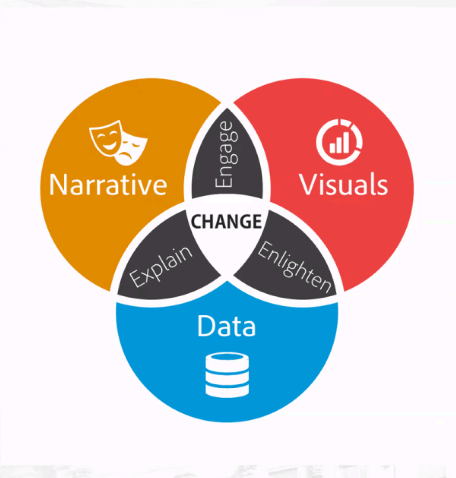
Data Context is the complete, operational understanding of data—covering what the data means, where it comes from, how it changes, whether it can be trusted, and how it should be used.

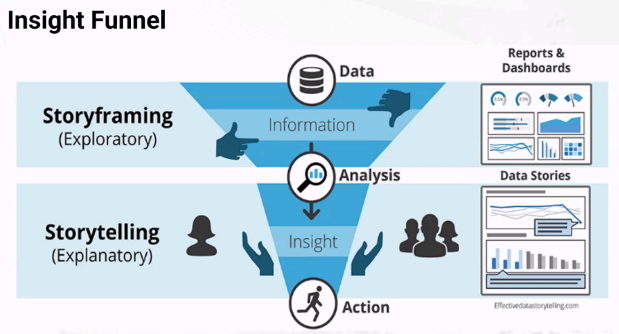
In practical terms, Data Context answers five fundamental questions:

- **Meaning** – What does this data represent in business terms? (metadata, data defined by data)
- **Origin** – Where did this data come from?
- **Transformation** – How has it been transformed across pipelines?
- **Trust** – Is this data accurate, fresh, and reliable right now?
- **Usage** – Who can use it, for what purpose, and under which constraints?

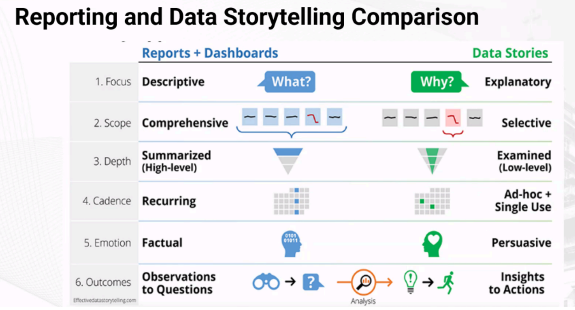
What Is a Data Story?

- A data story connects: **problem, evidence, meaning, and action.**
- **Data Story** is an evidence-based narrative that explains a meaningful situation through data and analysis and communicates its implications for understanding, decision, or action.
- **Data Storytelling** is the purposeful integration of evidence, analysis, visualization, context, and narrative to communicate meaning to a specific audience.





There is to consider meaningless stories that one should avoid (ex. Weather patterns to chicken wings business inside a mall, where rental costs, utilities, and nearest competitors are the main relevant factors)



Consider this

- Low sales, and you think it's bad (what if it was intentional due to renovations)
- Chicken wing business needs aircon just because February-March sales (what if it was just that the staff are on leave or vacation?)

Is visualization = data storytelling?

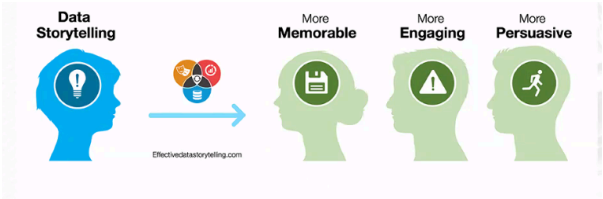
NO. Data visualization is not yet data storytelling.

- **Data Visualization** is the representation of data through graphical or visual forms to support exploration, understanding, comparison, and communication.
- **Exploratory Visualization** helps analysts discover possible patterns.
- **Explanatory Visualization** helps communicate a specific finding or message to an audience.

What, then, is data storytelling?

Data storytelling is the broader skill of communicating insights in a way that resonates with your audience.

This technique brings together data analysis, visualization, domain knowledge, and thoughtful presentation to make insights clear, compelling, and actionable. Instead of just sharing data, you deliver a data story with clarity, context, and purpose, so your message resonates.

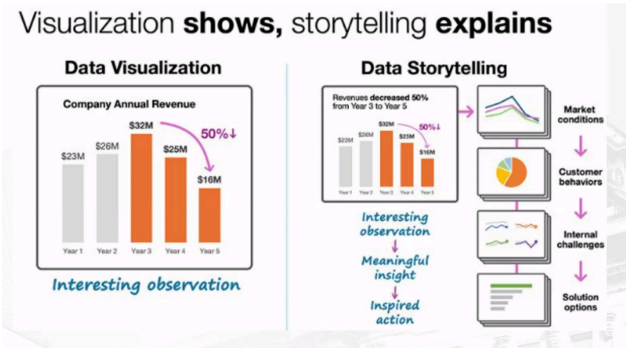


What isn't data storytelling?

- **Descriptive commentary.** Some people view data storytelling as simply commentary that accompanies charts that describes the information being displayed. Data storytelling extends well beyond just describing data—*the what*—and delves into *the why* to

enlighten people's minds. While calling out unusual anomalies, patterns, or trends in the data may be interesting, you have an incomplete data story if you can't explain why they occurred.

- **Added context.** Context is essential in data storytelling as it can provide greater clarity into what's happening. Having more background information is a step up from simply describing the data. However, merely providing more context doesn't mean you have created a data story. Stories follow a narrative arc or structure where events are revealed in a specific sequence. Without a clear storyline, you don't have a data story even with rich amounts of context.
- **Data charts.** You'll often hear people say, "Every chart tells a story." However, this is simply not true. While some graphs may contain some interesting data and yield part of a data story, a single chart can rarely fully explain 'the why' behind an entire problem or opportunity. Frequently, you need multiple charts to dig into an issue and properly explain it. In addition, many charts are designed to aid in the exploration of the data, not the explanation of the data. We can't expect charts alone to tell data stories when they weren't intended for that purpose.
- **Data visualization (in general).** Data storytelling is often used as a synonym for data visualization, especially when it adheres to best practices. However, just because you're following data visualization best practices doesn't mean you're automatically telling a data story. Instead, your purpose might be to help people explore a particular data set, which isn't storytelling. In addition, visualization is only one aspect of a data story. While data stories can actually exist without visuals (e.g., an audio data story featured in a podcast), data storytelling cannot happen without narrative.
- **Dashboards.** Many people and analytics vendors continue to believe dashboards tell data stories. However, dashboards are primarily designed to help us monitor business performance and explore data sets. While dashboards can be useful in helping us identify potential insights, they can be equally cumbersome and insufficient for curating and telling focused, persuasive data stories. While many analytics platforms offer some automated storytelling capabilities, this technology is limited to only *describing* anomalies in the data, not *explaining* what caused them.



How do we allow visualization to have context to the audiences that we have?

How do you make a story for data storytelling?



Analysis vs. Analytics

Analysis: What does the data show?

- Analysis is the systematic examination of data or evidence to identify patterns, relationships, differences, explanations, or estimates.

Analytics: What decision does it improve?

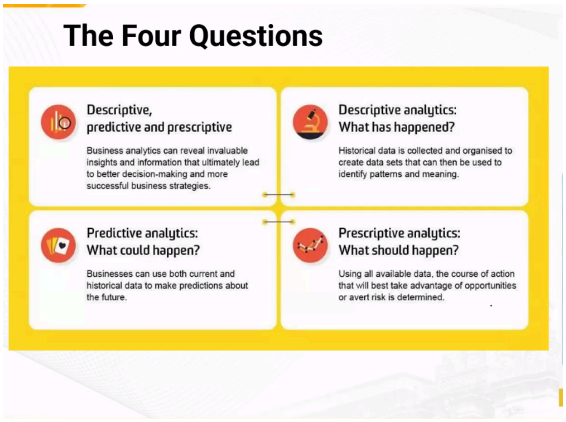
- Analytics is broader because it connects analysis to a decision or action.

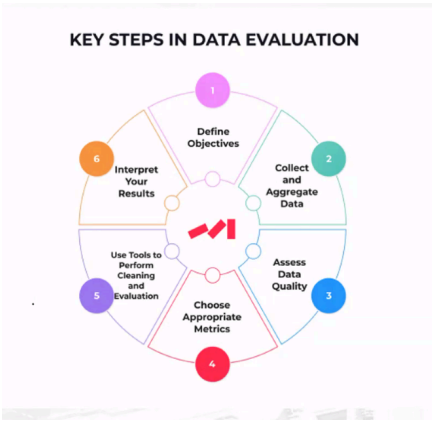
DATA ANALYSIS VS DATA ANALYTICS (Table Graphic)

Data Analysis	**Data Analytics**
Data Definition, Cleaning, Investigation And Transformation Into Meaningful Results.	Data Collection & Its Inspection
Used In Business To Analyze Data And Extract Useful Insights From The Data	Used In Businesses To Make Verdicts From Data Which Are Data-Driven
Used To Perform Predictive Analysis, Descriptive Analysis, Exploratory Analysis, Inferential Analysis	Used To Find Market Trends, Customer Preferences, Masked Patterns, Anonymous Correlations
RapidMiner, KNIME, Google Fusion Tables, Tableau Public, NodeXL, WolframAlpha Are Used For Data Analysis	For Data Analytical Purpose Python, SAS, R, Tableau Public, Apache Spark, Excel Are Frequently Used

The Four Questions

- **Descriptive Analytics** summarizes past or current data to describe events, conditions, performance, and patterns. (What happened?)
- **Diagnostic Analytics** investigates relationships, comparisons, and contextual factors to understand why an observed pattern may have occurred. (Why did it happen?)
- **Predictive Analytics** uses historical and current evidence to estimate future, unknown, or unobserved outcomes. (What might happen next?)
- **Prescriptive Analytics** uses data, objectives, models, rules, and constraints to recommend or evaluate possible actions. (What should we do?)





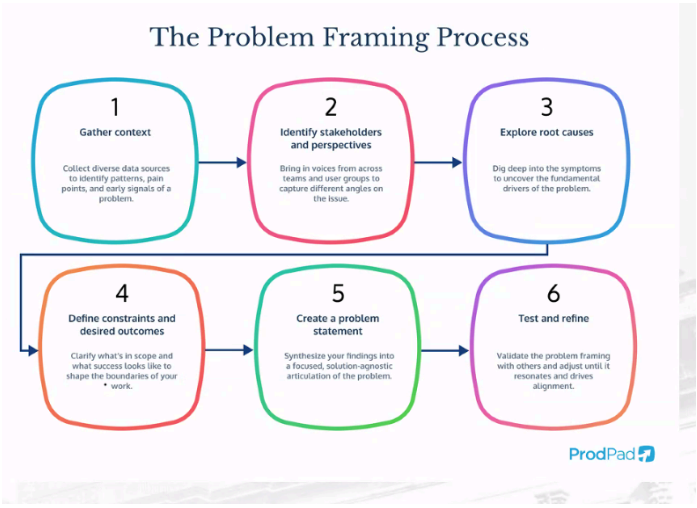


Problem Before Tool

- **Symptom** is an observable sign that something may need attention.
- **Problem** is the underlying condition, need, risk, or opportunity requiring understanding or action.
- **Problem Framing** is the process of defining the real problem, why it matters, who is affected, what constraints exist, and what successful improvement would mean.

The Missing Question

- A dashboard is not the ending. A model is not the ending. Impact is the ending.
- **Evaluation** is the systematic assessment of whether an analytics solution, decision, or action achieved its intended purpose.
- **Impact** is the meaningful change produced, supported, or influenced by an action or intervention.



Whose Story Is This?

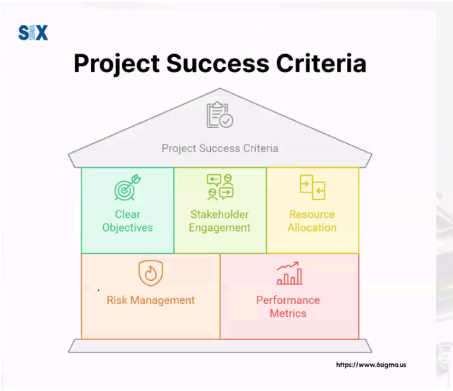
- **Stakeholder** is a person or group affected by, responsible for, or interested in the analytics solution.

- **Decision-maker** is the person or group responsible for choosing or authorizing an action.
- **Beneficiary** is the person or group intended to benefit from the decision or intervention.



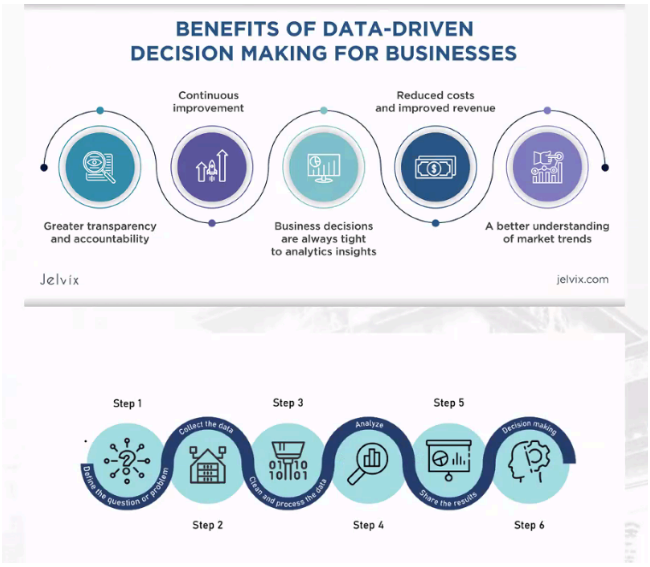
Objectives and Limits

- What do we want to improve?
- What limits what we can do?
- **Objective** is the desired result or improvement.
- **Constraint** is a limitation that restricts possible actions.
- **Success Criterion** is the condition used to judge whether the objective was achieved.



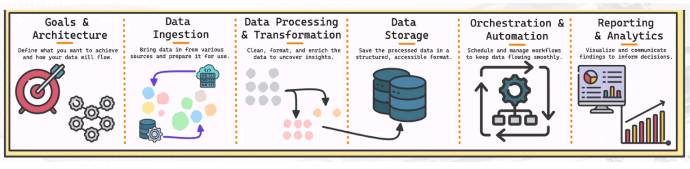
Decision-Driven Analytics

- **Decision-Driven Analytics** begins by identifying the decision that must improve, then determines what data, analysis, and tools are needed.
- **Problem-Driven Analytics** begins with the real-world problem, objective, and context before selecting methods or technologies.
- **Tool-Driven Analytics** begins with a preferred tool, model, algorithm, or technology, then searches for a problem to justify using it.



The End-to-End Lifecycle

- **End-to-End Analytics** is the complete lifecycle through which a real-world need is framed as an analytics problem, supported with appropriate data and methods, translated into a usable solution, deployed into a decision process, and continuously evaluated.



End-to-End Data Story

Problem → Evidence → Insight → Decision → Action → Impact

- **End-to-End Data Story** is the complete evidence-based narrative of an analytics solution, from problem and context to data, analysis, insight, decision, action, impact, and learning.

